

ERIC ODOOM

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EDUCATION

Doctor of Philosophy, Statistics

Expected Dec. 2026

University of Cincinnati, Cincinnati, OH

GPA: 3.88/4.00

Dissertation: Scalable and Robust Bayesian Methods for Variable Selection in High-Dimensional and Incomplete Data

Advisor: Professor Xia Wang

Master of Science, Mathematics

Apr. 2021

Southeast Missouri State University, Cape Girardeau, MO

GPA: 3.92/4.00 | Thesis: Predicting Healthcare Expenditure: Perspectives from Statistical Learning and Machine Learning

Advisor: Professor Emmanuel Thompson

Bachelor of Science, Statistics & Mathematics -

May 2018

University of Ghana-Legon, Accra, Ghana

GPA: 3.64/4.00 | Thesis: Prediction of Ghana's Presidential Election Using the Dirichlet Distribution via Bayesian Methodology

Advisor: Dr. Ezekiel Nii Noye Nortey

RESEARCH INTERESTS

Bayesian computation; high-dimensional statistics; statistical learning; variable selection; robust statistical inference; uncertainty quantification; missing-data methods; predictive modeling.

PUBLICATIONS & MANUSCRIPTS

Publications

- Odoom, E., Ouyang, J., Cao, X., & Wang, X. (2026). Hierarchical skinny Gibbs sampler in logistic regression using Pólya-Gamma latent variables. *Statistics and Its Interface*, 19(2), 179-196.
- Odoom, E., & Wang, X. (2026). Consistent and scalable variable selection with robust link functions. *The New England Journal of Statistics & Data Science*, 1-15. doi:10.51387/26-NEJSDS102
- Knodell, M., Dean, A., Kroger, L., McCreary, T., Odoom, E., Earl, B., & Zhang, F. (2026). Early signs of aging in auditory function reflected in the event-related potentials. *Journal of the American Academy of Audiology*, 37(3), 209-221. doi:10.3766/jaaa.250015

Manuscripts Under Review

- Fagbohunbe, H. T., Odoom, E., Wang, X., & Cao, X. Bayesian automated learning of sparsity in risk prediction with application to whole-brain functional connectivity analysis. Submitted.
- Odoom, E., Wang, X., Beebe, W. E., & Zhang, N. Bayesian Dirichlet regression for correlated compositional outcomes with application to an experimental sleep study. Submitted.

RESEARCH EXPERIENCE

Doctoral Researcher, Statistics - University of Cincinnati

Ph.D. Research

- Develop scalable and robust Bayesian variable-selection methods for high-dimensional and incomplete data, including continuous and discrete spike-and-slab formulations.
- Derive Gibbs sampling algorithms using Pólya-Gamma data augmentation and hierarchical Bayesian models for high-dimensional logistic regression.
- Design simulation studies to evaluate variable-selection accuracy, robustness, and computational performance under complex missing-data mechanisms.
- Develop reproducible R implementations and investigate applications to high-dimensional biomedical and scientific data.

Statistician Intern - Cincinnati Children's Hospital

May 2026 - Present

Medical Center, Cincinnati, OH

Research Advisor: Dr. Nanhua Zhang

- Served as lead statistical analyst for proteomics and translational medicine studies, developing statistical frameworks for multi-omics and clinical data integration in biomarker discovery and precision medicine.
- Developed and applied statistical and machine-learning methods for high-dimensional proteomic and clinical data in idiopathic pulmonary fibrosis, interstitial lung diseases, and related pulmonary disorders.

Research Assistant (Intern) - Cincinnati Children's

May 2024-Aug. 2025

Hospital Medical Center, Cincinnati, OH

Research Advisors: Dr. Nanhua Zhang and Professor Xia Wang

- Collaborated with clinicians and multidisciplinary research teams to analyze high-dimensional proteomic and auditory data, applying statistical modeling to identify biomarkers associated with disease progression and age-related functional decline.
- Developed reproducible analytical pipelines and Bayesian models for longitudinal, compositional, and time-to-event data.

- Designed and deployed an R Shiny application integrated with Qualtrics for real-time data ingestion, automated analysis, and dynamic reporting.

Statistician Intern - Procter & Gamble / University of Cincinnati Simulation Center, Cincinnati, OH

Aug. 2021-Aug. 2024

- Developed and deployed R Shiny analytical applications supporting experimental design, statistical modeling, and product performance evaluation.
- Applied statistical modeling and experimental analysis to product efficacy studies and translated results for research, development, and business stakeholders.
- Modernized legacy SAS analytical tools into validated, reproducible R-based platforms and partnered with cross-functional teams on statistical requirements and implementation.

SELECTED PRESENTATIONS

- Invited Talk - 39th New England Statistics Symposium, Storrs, CT (May 2026): Consistent and scalable variable selection with robust link functions.
- University of Cincinnati Statistics Seminar (Spring 2026): Consistent and scalable variable selection with robust link functions.
- Joint Statistical Meetings (JSM), Nashville, TN (Aug. 2025): Hierarchical skinny Gibbs sampler in logistic regression using Pólya-Gamma latent variables.
- ENAR Spring Meeting, New Orleans, LA (Mar. 2025): Hierarchical skinny Gibbs sampler in logistic regression using Pólya-Gamma latent variables.
- ASA Kentucky Chapter Spring Meeting, Lexington, KY (Mar. 2025): Hierarchical skinny Gibbs sampler in logistic regression using Pólya-Gamma latent variables.
- Joint Statistical Meetings (JSM), Portland, OR (Aug. 2024): Bayesian Dirichlet regression for correlated compositional outcomes with application to an experimental sleep study.

TEACHING EXPERIENCE

Graduate Teaching Assistant - Southeast Missouri State University, Cape Girardeau, MO

Aug. 2019 - Apr. 2021

- Designed individualized tutoring approaches for students with varied backgrounds, increasing the proportion earning at least a B in Introduction to Statistics from 50% to 90%.
- Assisted with instruction, grading, tutoring, review materials, and group study sessions for statistics, precalculus, calculus, statistical reasoning, and algebra.

Teaching Assistant (National Service Personnel) - University of Ghana-Legon, Accra, Ghana

Aug. 2018 - Jul. 2019

- Led tutorials and examination review sessions in statistical inference, business statistics, and introductory statistics.

FELLOWSHIPS, AWARDS & HONORS

- Taft Dissertation Completion Fellowship Award, University of Cincinnati (Aug. 2025-Apr. 2026)
- Second Place Award for Oral Presentation, ASA Kentucky Chapter Spring Meeting (Mar. 2025)
- Taft Graduate Student Travel Award (Aug. 2025; Aug. 2024)
- Graduate Student Government Conference Travel Award (Mar. 2025; Aug. 2024; Mar. 2024)
- Russel and Elnora Michel Mathematics Prize Award (Aug. 2020)

SKILLS & TECHNICAL COMPETENCIES

Statistical Methodology: Bayesian inference and computation, high-dimensional variable selection, spike-and-slab models, Pólya-Gamma data augmentation, generalized linear models, survival analysis, longitudinal and mixed-effects models, missing-data methods, statistical learning, predictive modeling, and experimental design.

Computing: R (advanced), Python, SAS, SQL, R Shiny, Git; statistical simulation, reproducible research, data visualization, data wrangling, and large-scale data analysis.

Research & Communication: Scientific writing, interdisciplinary collaboration, presentation of statistical methodology and applied research, reproducible workflows, and research integrity.

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA), International Biometric Society (IBS), Eastern North American Region (ENAR), Society for Industrial and Applied Mathematics (SIAM), American Mathematical Society (AMS), Ghana Statistical Society (GSS).